

Test Model on Multiple Datasets

This notebook evaluates a trained model on three test datasets:

- **Natural:** Uniformly random samples from $[1, 10^{13}]$
- **Cheat:** Numbers with prime factors only within the first 100 primes
- **Non-cheat:** Numbers with at least one prime factor outside the first 100 primes

It reports per-class performance for each dataset.

Configuration

Specify the model checkpoint and encoding to test.

IMPORTANT:

- `ENCODING` should be the **base encoding ONLY** (e.g., `interCRT100`, `CRT100`)
- Do **NOT** include dataset type suffixes like `_natural`, `_cheat`, or `_non_cheat`
- The notebook will automatically test on all three dataset types

Device detection: GPU not available

Using device: cpu

Testing model: `../models/model_interCRT100_natural/mu/1/best-valid_arithmetic_acc.pth`

Encoding: `interCRT100`

Task: `mu`

Results will be saved to: `../test_results/interCRT100_mu`

Helper Functions

Helper functions loaded!

Check if Model Checkpoint Exists

✓ Checkpoint found: `../models/model_interCRT100_natural/mu/1/best-valid_arithmetic_acc.pth`

Check Test Data Files

Checking test data files:

```
✓ natural      : ../input/input_dir_interCRT100_natural/mu_interCRT100_natural.txt.test (105.83 MB)
✓ cheat        : ../input/input_dir_interCRT100_cheat/mu_interCRT100_cheat.txt.test (105.59 MB)
✓ non_cheat    : ../input/input_dir_interCRT100_non_cheat/mu_interCRT100_non_cheat.txt.test (105.82 MB)
```

Run Evaluation on Each Test Dataset

Evaluation function ready!

Evaluate on Natural Dataset

```
=====
EVALUATING ON NATURAL DATASET
=====
```

Running in CPU mode

Running command:

```
python ../Int2Int/train.py --eval_only True --reload_model /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/./models/model_interCRT100_natural/mu/1/best-valid_arithmetic_acc.pth --eval_data /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/./input/input_dir_interCRT100_natural/mu_interCRT100_natural.txt.test --eval_size 10000 --data_types int[200]:range(-1,2) --operation data --num_workers...
```

```
=====
Full output saved to: ../test_results/interCRT100_mu/eval_natural.log
```

✓ Evaluation completed successfully!

Overall Accuracy: 52.23%

Evaluate on Cheat Dataset

EVALUATING ON CHEAT DATASET

Running in CPU mode

Running command:

```
python ../Int2Int/train.py --eval_only True --reload_model /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/./models/model_interCRT100_natural/mu/1/best-valid_arithmetic_acc.pth --eval_data /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/./input/input_dir_interCRT100_cheat/mu_interCRT100_cheat.txt.test --eval_size 10000 --data_types int[200]:range(-1,2) --operation data --num_workers...
```

Full output saved to: ../test_results/interCRT100_mu/eval_cheat.log

✓ Evaluation completed successfully!

Overall Accuracy: 42.54%

Evaluate on Non-Cheat Dataset

EVALUATING ON NON-CHEAT DATASET

Running in CPU mode

Running command:

```
python ../Int2Int/train.py --eval_only True --reload_model /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/./models/model_interCRT100_natural/mu/1/best-valid_arithmetic_acc.pth --eval_data /Users/clairexu/Desktop/GitHub/mobius_case_study/notebooks/./input/input_dir_interCRT100_non_cheat/mu_interCRT100_non_cheat.txt.test --eval_size 10000 --data_types int[200]:range(-1,2) --operation data --num_workers...
```

Full output saved to: ../test_results/interCRT100_mu/eval_non_cheat.log

✓ Evaluation completed successfully!

Overall Accuracy: 52.00%

Compile Results

✓ Successfully evaluated on 3 dataset(s)

Results saved to: ../test_results/interCRT100_mu/all_results.json

Overall Performance Comparison

OVERALL PERFORMANCE SUMMARY

	Dataset	Accuracy (%)	Acc $\mu=0$ (%)	Acc $\mu=1$ (%)	Acc $\mu=-1$ (%)	Perfect (%)	Correct (%)	XE Loss
0	Natural	52.23	89.532806	17.307061	39.137591	52.23	52.23	0.944391
1	541-smooth	42.54	45.812374	12.567132	44.225604	42.54	42.54	1.777523
2	non-541-smooth	52.00	90.277065	15.533981	39.873838	52.00	52.00	0.932312

Summary saved to: `../test_results/interCRT100_mu/overall_summary.csv`

Precision, Recall, F1, and Confusion Matrix

Note: The Int2Int evaluator provides per-class accuracy but not the full confusion matrix. To compute precision, recall, and F1 properly, we need to run a custom evaluation that captures predictions vs ground truth.

The following section runs a custom evaluation to extract predictions and compute these metrics.

Classification metrics and parsing functions loaded!

Compute Classification Metrics from Evaluation Results

Use the predictions captured from the evaluation runs above:

=====
PARSING PREDICTIONS: NATURAL
=====

Parsing predictions from: ../test_results/interCRT100_mu/eval_dump/natural/debug/o1nf4ugwdy/eval.valid.arithmetic.0
Extracted 10000 predictions
Metrics computed successfully!

=====
PARSING PREDICTIONS: CHEAT
=====

Parsing predictions from: ../test_results/interCRT100_mu/eval_dump/cheat/debug/0ttpmtvs7j/eval.valid.arithmetic.0
Extracted 10000 predictions
Metrics computed successfully!

=====
PARSING PREDICTIONS: NON_CHEAT
=====

Parsing predictions from: ../test_results/interCRT100_mu/eval_dump/non_cheat/debug/wg5d7fygnz/eval.valid.arithmetic.0
Extracted 10000 predictions
Metrics computed successfully!

=====
Classification metrics computed for 3 dataset(s)
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Classification Report: Precision, Recall, F1

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CLASSIFICATION REPORT: Natural

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Samples evaluated: 10000

Invalid predictions: 0

	precision	recall	f1-score	support
$\mu=-1$	0.48	0.39	0.43	3038
$\mu=0$	0.56	0.90	0.69	3917
$\mu=1$	0.41	0.17	0.24	3045
accuracy			0.52	10000
macro avg	0.49	0.49	0.46	10000
weighted avg	0.49	0.52	0.48	10000

Summary Metrics:

Metric	Macro	Weighted
Precision	0.4858	0.4922
Recall	0.4866	0.5223
F1-Score	0.4551	0.4755

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CLASSIFICATION REPORT: 541-smooth

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Samples evaluated: 10000

Invalid predictions: 0

	precision	recall	f1-score	support
$\mu=-1$	0.12	0.44	0.19	1117
$\mu=0$	0.83	0.46	0.59	7952
$\mu=1$	0.08	0.13	0.09	931
accuracy			0.43	10000
macro avg	0.34	0.34	0.29	10000
weighted avg	0.68	0.43	0.50	10000

Summary Metrics:

Metric	Macro	Weighted
Precision	0.3431	0.6826
Recall	0.3420	0.4254
F1-Score	0.2919	0.5000

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CLASSIFICATION REPORT: non-541-smooth

Samples evaluated: 10000
Invalid predictions: 0

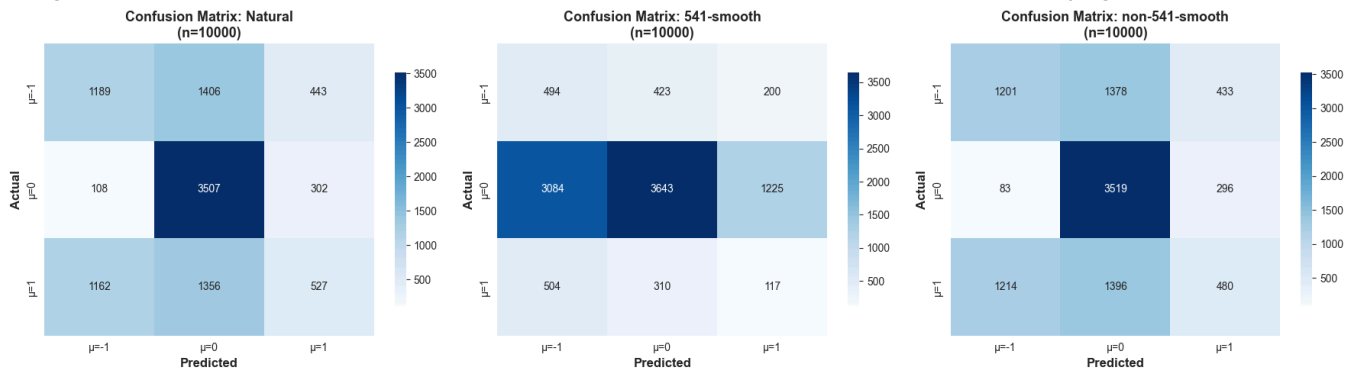
	precision	recall	f1-score	support
$\mu=-1$	0.48	0.40	0.44	3012
$\mu=0$	0.56	0.90	0.69	3898
$\mu=1$	0.40	0.16	0.22	3090
accuracy			0.52	10000
macro avg	0.48	0.49	0.45	10000
weighted avg	0.49	0.52	0.47	10000

Summary Metrics:

Metric	Macro	Weighted
Precision	0.4790	0.4855
Recall	0.4856	0.5200
F1-Score	0.4500	0.4695

Confusion Matrices

Figure saved to: ../test_results/interCRT100_mu/confusion_matrices.png



Per-Class Precision, Recall, F1

PER-CLASS METRICS: Natural

	Class	Precision	Recall	F1-Score
0	$\mu=-1$	0.4835	0.3914	0.4326
1	$\mu=0$	0.5594	0.8953	0.6886
2	$\mu=1$	0.4143	0.1731	0.2442

Saved to: ../test_results/interCRT100_mu/per_class_metrics_natural.csv

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PER-CLASS METRICS: 541-smooth

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	Class	Precision	Recall	F1-Score
0	$\mu=-1$	0.1210	0.4423	0.1900
1	$\mu=0$	0.8325	0.4581	0.5910
2	$\mu=1$	0.0759	0.1257	0.0946

Saved to: ../test_results/interCRT100_mu/per_class_metrics_cheat.csv

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PER-CLASS METRICS: non-541-smooth

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	Class	Precision	Recall	F1-Score
0	$\mu=-1$	0.4808	0.3987	0.4359
1	$\mu=0$	0.5592	0.9028	0.6906
2	$\mu=1$	0.3970	0.1553	0.2233

Saved to: ../test_results/interCRT100_mu/per_class_metrics_non_cheat.csv

Save All Detailed Metrics

All detailed metrics saved to: ../test_results/interCRT100_mu/detailed_classification_metrics.json

Predictions saved to: ../test_results/interCRT100_mu/predictions_natural.npz

Predictions saved to: ../test_results/interCRT100_mu/predictions_cheat.npz

Predictions saved to: ../test_results/interCRT100_mu/predictions_non_cheat.npz

Manual Inspection: Sample Predictions from Each Test Set

Randomly select 20 samples from each test dataset and display the number, true Mobius value, and model prediction.

natural: loaded 10000 numbers

cheat: loaded 10000 numbers

non_cheat: loaded 10000 numbers

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SAMPLE PREDICTIONS FROM EACH TEST SET

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DATASET: Natural (showing 20 random samples)

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Correct: 11/20 (55.0%)

	n	True $\mu(n)$	Predicted $\mu(n)$	Correct
0	(sample 84)	0	0	Yes
1	(sample 733)	1	0	No
2	(sample 966)	1	-1	No
3	(sample 1359)	-1	-1	Yes
4	(sample 1485)	0	0	Yes
5	(sample 2085)	-1	0	No
6	(sample 2550)	0	0	Yes
7	(sample 2885)	0	0	Yes
8	(sample 4742)	0	0	Yes
9	(sample 5540)	1	-1	No
10	(sample 6247)	1	1	Yes
11	(sample 7654)	1	1	Yes
12	(sample 8066)	1	0	No
13	(sample 8795)	0	0	Yes
14	(sample 8886)	1	1	Yes
15	(sample 8999)	1	0	No
16	(sample 9110)	-1	0	No
17	(sample 9298)	-1	0	No
18	(sample 9424)	0	0	Yes
19	(sample 9573)	1	0	No

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DATASET: 541-smooth (showing 20 random samples)

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Correct: 9/20 (45.0%)

	n	True $\mu(n)$	Predicted $\mu(n)$	Correct
0	(sample 28)	0	-1	No
1	(sample 80)	-1	0	No
2	(sample 221)	0	1	No
3	(sample 343)	0	0	Yes
4	(sample 582)	0	-1	No
5	(sample 1123)	0	0	Yes
6	(sample 1134)	0	1	No
7	(sample 1281)	-1	-1	Yes
8	(sample 2123)	0	-1	No
9	(sample 2561)	-1	-1	Yes
10	(sample 4567)	-1	-1	Yes
11	(sample 5091)	0	0	Yes
12	(sample 5129)	0	0	Yes
13	(sample 5421)	0	-1	No
14	(sample 7757)	0	0	Yes
15	(sample 7922)	0	0	Yes
16	(sample 8327)	0	1	No
17	(sample 8528)	0	-1	No
18	(sample 9763)	0	1	No
19	(sample 9994)	0	-1	No

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DATASET: non-541-smooth (showing 20 random samples)

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Correct: 9/20 (45.0%)

	n	True $\mu(n)$	Predicted $\mu(n)$	Correct
0	(sample 2206)	0	0	Yes
1	(sample 2251)	-1	0	No
2	(sample 2876)	1	-1	No
3	(sample 3256)	1	-1	No
4	(sample 3620)	0	0	Yes
5	(sample 3707)	0	0	Yes
6	(sample 4117)	0	0	Yes
7	(sample 5251)	1	-1	No
8	(sample 5884)	1	0	No
9	(sample 6020)	0	0	Yes
10	(sample 6760)	0	0	Yes
11	(sample 6794)	-1	0	No
12	(sample 7247)	1	1	Yes
13	(sample 7279)	-1	-1	Yes
14	(sample 7686)	0	0	Yes
15	(sample 7882)	-1	1	No
16	(sample 8065)	-1	0	No
17	(sample 8365)	0	1	No
18	(sample 8421)	0	1	No
19	(sample 9057)	-1	0	No

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Square-Free Analysis

Analyze the distribution of Möbius function values based on whether numbers are square-free or not.

Note: This analysis is only available for encodings that include the square-free flag (e.g., `n_interCRT_sqfree_mobius`).

Parsing natural test file...

△ No sqfree flag found in encoding – deriving square-free info from $\mu(n)$.

Total samples: 10000

Square-free: 6083 (60.8%)

Non-square-free: 3917 (39.2%)

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SQUARE-FREE NUMBERS ($\mu(n) \in \{-1, 1\}$)

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True distribution:

$\mu=-1$: 3038 (49.94%)

$\mu=0$: 0 (0.00%)

$\mu=1$: 3045 (50.06%)

Predicted distribution:

$\mu=-1$: 2351 (38.65%)

$\mu=0$: 2762 (45.41%)

$\mu=1$: 970 (15.95%)

Accuracy: 28.21%

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NON-SQUARE-FREE NUMBERS ($\mu(n)$ must be 0)

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True distribution:

$\mu=-1$: 0 (0.00%)

$\mu=0$: 3917 (100.00%)

$\mu=1$: 0 (0.00%)

Predicted distribution:

$\mu=-1$: 108 (2.76%)

$\mu=0$: 3507 (89.53%)

$\mu=1$: 302 (7.71%)

Accuracy: 89.53%

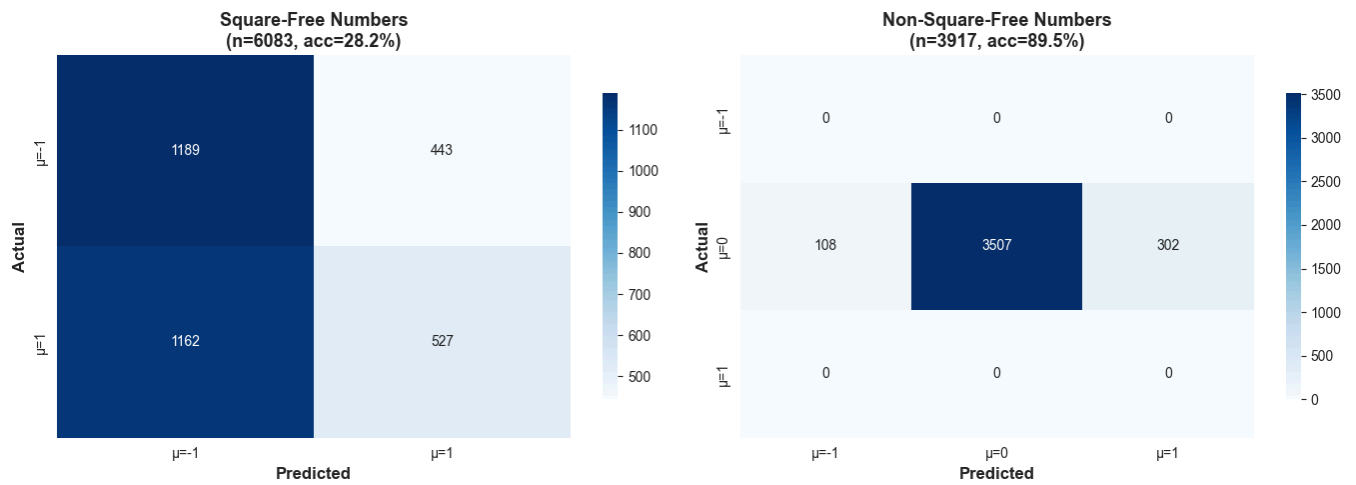
△ Model Error: 410 non-square-free predicted with $\mu \neq 0$

Constraint accuracy: 89.53%

Square-Free Confusion Matrices

Visualize the confusion matrices separately for square-free and non-square-free numbers.

Figure saved to: ../test_results/interCRT100_mu/confusion_matrices_square_free.png



CONFUSION MATRIX SUMMARY

Square-free numbers (6083 samples):
 Correct predictions: 1716
 Accuracy: 28.21%

Non-square-free numbers (3917 samples):
 Correct predictions ($\mu=0$): 3507
 Accuracy: 89.53%
 Violations (predicted $\mu \neq 0$): 410